

CONTACT

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San Francisco Bay Area

TECHNICAL SKILLS

Spark Hive Hadoop Databricks
Delta Lake Snowflake Redshift dbt
Airflow Dataswarm AWS Azure
GCP Terraform Kubernetes Python
SQL Scala LLM/RAG Pipelines
Claude Code AI Agents Prompt Eng.
LangChain GitHub Copilot MLflow
Diff. Privacy Unity Catalog Data Mesh

EDUCATION

Data Engineering Cert.
UC Santa Cruz, 2013–14
B.S. Electrical Engineering
Sardar Patel College of Eng.
Diploma, Electronics & Comms
Technical Board, Tamil Nadu

PUBLICATIONS

- Cloud Computing
- All About Big Data
- Future Trends in BI

AI & THOUGHT LEADERSHIP

- 160+ articles on paddyspeaks.com
- Claude Code skill development
- Vibe coding methodology
- AI agent architecture writing

OPEN-SOURCE & COMMUNITY

- Interview Studio — free Data & AI Engineering prep
- 22 data-model deep-dives + 1,500+ questions
- In-browser SQL / Python (WASM), no backend

EARLIER CAREER

Data Engineer Consultant

LinkedIn
Dec 2017 – Apr 2018
GDPR compliance — PII encryption into Dali storage (Hive, Python, Pig)

Lead Data Engineer

Meta
Sep 2015 – Sep 2017
80+ Dataswarm pipelines for petabyte-scale ads; Cross Device Insights, Outcomes Datamart + Norms DB

Data Engineer

GREE International
2014 – 2015
PII masking, Vertica → Redshift, 1,000+ table optimization

Data Architect

Chegg Inc.
Oct 2013 – Apr 2014

Technical Director

Model N
2011 – 2013
Life Sciences BI, cloud migration, ETL modularization

Architect

Paddy Iyer

DATA ENGINEERING LEADER · STRATEGIC DATA ARCHITECT · AI & PRIVACY ENGINEERING · AI INNOVATOR

15+ years driving enterprise data transformation for global tech leaders across cloud/SaaS, fintech, gaming, and consumer platforms. Expert in cloud-native architectures, real-time analytics, privacy-first pipelines, and AI-augmented data operations. Proven builder of high-trust, high-performance data teams at petabyte scale. Published writer and thought leader on data architecture, AI agents, and the intersection of ancient wisdom with modern technology.

Modern Data Architecture Privacy-First Design Consent Architecture AI Agents & LLM Pipelines Prompt Engineering Data Governance (C1–C5) Petabyte-Scale Systems Global Team Leadership Data Storytelling

FEATURED EXPERIENCE

Privacy Data Engineering *Apr 2024 – Present*

Meta | Safe Ads Program

Spearheading Meta's transformation from broad data access to consent-first advertising via the Safe Ads Program, remediating 700+ data assets across AAP, Consent Revocation (CR), BYOU, and ICOU privacy domains

Data Architecture & Privacy Remediation

- Architected an end-to-end 8-stage privacy remediation pipeline (classification → lineage → topology sort → pipeline dev → diff → deploy → compliance → downstream migration) supporting four parallel privacy workstreams: AAP, CR filtering, BYOU, and ICOU
- Achieved 100% remediation of all high-risk assets across AAP and Consent Revocation, directly enabling Meta's Safe Ads launch timeline
- Designed and deployed consent filtering pipelines for Instagram and Facebook tables — enforcing user-level opt-out at the data layer across BYOU and ICOU use cases
- Built signal table architectures and WaitForHive/WaitForOperator patterns for multi-partition compliance, replacing fragile direct-dependency models with auditable, consent-aware orchestration
- Designed DP anonymization operators for safe analytics, preserving analytical capability while respecting consent boundaries
- Landed 800+ code changes and managed 1,000+ tasks; reduced average asset remediation cycle time from weeks to days
- Closed privacy gaps across cross-functional Dataswarm pipelines involving 3+ producer teams, preventing compliance audit failures

AI-Augmented Remediation — Claude Code Skill

- Co-developed with software engineers a custom “Claude Code Skill” (YAML + Markdown playbook) encoding the full AAP/CR/BYOU/ICOU remediation methodology — reduced new-member ramp-up from ~2 weeks to ~2 days
- Partnered with SWEs to augment AI implementation capabilities, enabling the skill to auto-generate pipeline codemods across all four privacy domains: DevmateHiveCustomPipeline transforms, consent filter generation, signal table wiring, and multi-producer table alignment
- Cut remediation iteration time by 3–5×: the AI agent handled boilerplate codemod generation, lint/test cycles, and cross-team dependency resolution — enabling sustained 800+ diff velocity
- Applied prompt engineering and LLM-assisted code analysis to reduce documentation overhead by ~70% and eliminate fragile metastore tracking

DataSwarm Pipeline Engineering

- Streamlined schema/dataset definitions, removed hardcoded paths, and standardized lint/schedule/test workflows — reducing production incidents
- Documented repeatable validation steps that cut time-to-first-deploy for new pipelines by ~50%
- Resolved production failures (multi-partition wait conditions, signal table dependencies) across Chronos-orchestrated Instagram and Facebook pipelines
- Applied SQL/Spark-style ETL best practices sustaining 800+ change velocity over the engagement

CallidusCloud

2005 – 2011

Incentive comp analytics, BusinessObjects
XI

DW Architect

Hewlett Packard

2001 – 2005

Enterprise DW + 8 datamarts; ETL 18 → 3
hrs; 40–50% sales lift via clickstream

EARLIER CAREER

Lead Data Architect — Partner Data Engineering *Jan 2022 – Jan 2024* VMware

Strategic architect for VMware's Partner Data Platform — unifying fragmented partner data into a single governed analytics layer

- Designed and built the Partner Data Platform from the ground up, unifying partner data from 10+ disparate sources into a single governed analytics layer — eliminating silos that had blocked cross-functional decision-making for years
- Implemented VMware's first structured data governance for partner data (classification, lineage, policy enforcement) plus a self-service data catalog — cutting tribal-knowledge reliance by 60% and ad-hoc request volume by ~40%, and achieving audit-ready status
- Pioneered an AI copilot for retrieval optimization, coding, and governance — one of VMware's earliest LLM-assisted data engineering deployments, reducing routine query development time by ~30%
- Achieved 40%+ Spark processing improvements via broadcast joins, caching, and skew management — bringing SLA-critical reports inside their delivery windows for the first time
- Engaged director/VP-level stakeholders to define a unified data vision, securing budget and headcount that grew the team from 3 to 8 engineers during the engagement

Senior Consultant — Ads, Commerce & Privacy *Jul 2019 – Dec 2021* Meta

Led high-impact projects across Meta's Ads, Commerce, and Privacy teams, architecting secure, scalable, and compliant data platforms

- Led privacy remediation across 1,200+ SMB 2.0, Customer Journey, and BPO tables — resolving critical compliance-risk design flaws and establishing remediation patterns later adopted at scale in the Safe Ads program
- Spearheaded migration of 1,000+ Hive pipelines to Spark, automating ~70% of conversion tasks — compressing a projected 12-month migration to under 6 months
- Owned the SMB funnel redesign, improving query performance 3x and scaling the pipeline from thousands to tens of thousands of small-business advertisers without infrastructure changes
- Enhanced Hive anonymization processes with privacy engineers, enabling ads and commerce data products to pass privacy audits without manual intervention — cutting audit preparation effort by ~60%